

Reproducing the TV posterior-mean denoiser — cameraman, $\sigma = t = 20/256$, $m = 8000$ sweeps, every network trained for 250000 steps

Neither LPN route inverts anything to DENOISE: $\nabla G = (\nabla \psi)^{-1}$, so both deploy $\nabla \psi_\theta$. The prox_{J_2} panel asks a different question — does the prior LPN (Ours) REPORTS have the denoiser as its prox? For the direct fit those two objects are identical by construction.

clean



noisy input
22.50 dB vs clean



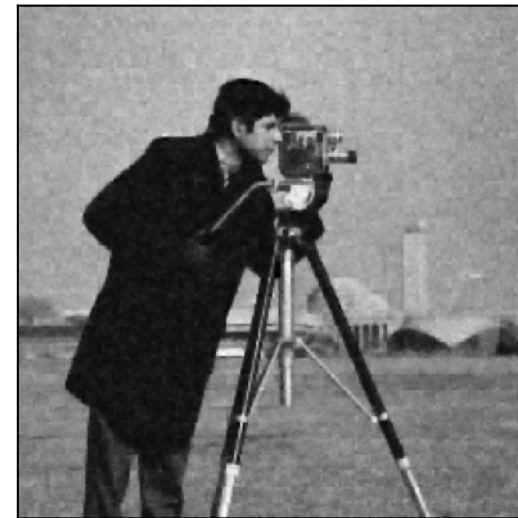
u_{PM} (MCMC sampler)
GROUND TRUTH
27.32 dB vs clean



$u_\theta = \nabla \psi_\theta(x)$
DEPLOYED by both LPN routes
26.92 dB vs clean



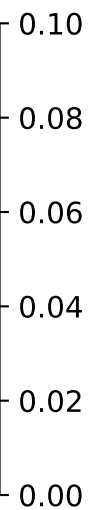
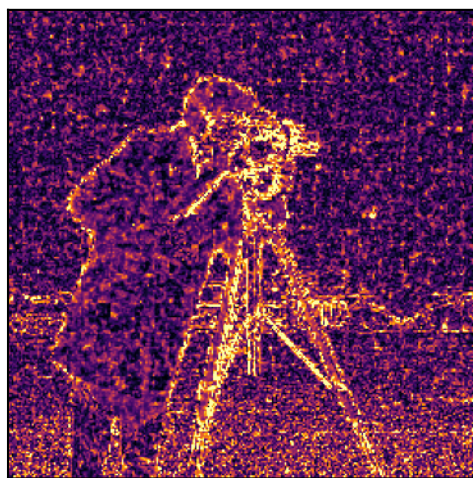
prox_{J_2} , LPN (Ours)
CHECK, not a denoiser
26.83 dB vs clean



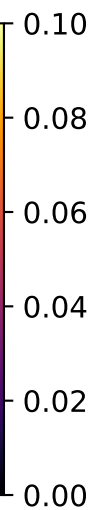
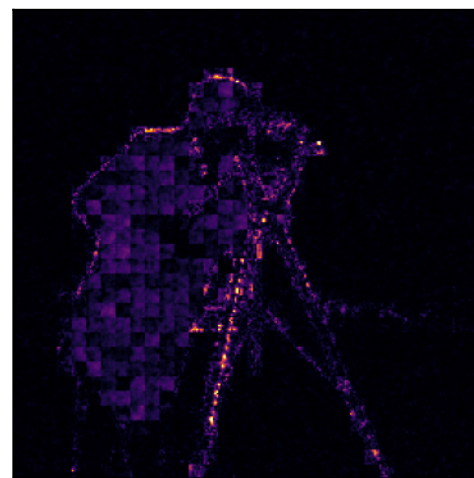
$\hat{u} = \text{prox}_{J_\theta}(x)$
DEPLOYED, direct fit
27.09 dB vs clean



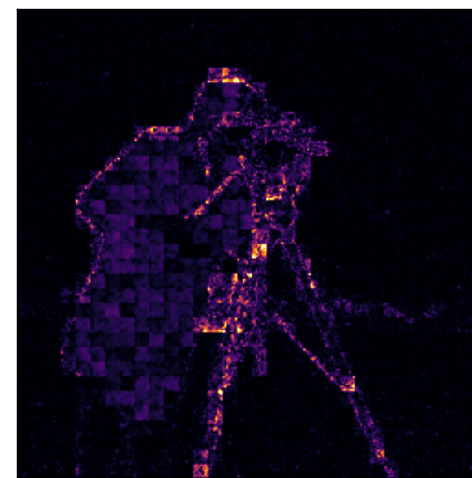
$|u_{PM} - \text{clean}|$



$|\nabla \psi_\theta - u_{PM}|$
41.76 dB vs u_{PM}



$|\text{prox}_{J_2} - u_{PM}|$
39.50 dB vs u_{PM}



$|\text{prox}_{J_\theta} - u_{PM}|$
46.93 dB vs u_{PM}

